

Snowflake Fundamentals — Student Setup Guide

Course: Snowflake Fundamentals 4-Day Class

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1. Overview

Complete this setup before Day 1. You need a Snowflake account, Snowflake CLI, a browser, and a local copy of the lab files. Snowflake CLI uses the `snow` command. SnowSQL is a legacy client and is not required for these labs.

Run commands beginning with `snow` in a local terminal (PowerShell, Command Prompt, or macOS/Linux Terminal). Run SQL through `snow sql` or in Snowsight, as directed. Local file transfers with PUT and GET use the CLI in this course; do not paste those commands into a Snowsight SQL editor.

2. Snowflake Account

2.1 Account access

Use the account supplied by the instructor, or create a trial at [Snowflake signup](#). Select Enterprise edition for the class features, and use the cloud/region specified by the instructor. Activate the account and confirm you can sign in to Snowsight.

2.2 Connection details

In Snowsight, open the account details and copy the account identifier in `organization-account` format. Do not use the entire browser URL as the CLI account value. Record your username and the class role and warehouse.

The labs use SYSADMIN and COMPUTE_WH where available. The instructor must ensure your user has the class role and that the role has warehouse access. Use administrative privileges only for setup steps that require them.

Confirm whether you will use a Snowflake password or the organization's SSO. Complete MFA when required by the account. Do not assume that a trial account has the same authentication policy as a corporate account.

3. Snowflake CLI Installation

For Windows, download **Snowflake CLI** from the [Snowflake CLI download page](#). Choose the Windows **MSI installer**, download it, and run it. Make sure you select Snowflake CLI, not SnowSQL. On managed computers, ask IT to install the tool before class.

After installation, open a **new Command Prompt window** and run:

```
snow --version
```

On the first run, you may see Unpacking distribution (tar.bz2). Wait for the preparation to finish and for the version number to appear. This initial preparation can take time; do not interrupt it.

Expected: the Snowflake CLI version number. For command help, run `snow --help`. For Windows upgrades, download and run the appropriate MSI installer; do not assume automatic upgrades.

For other operating systems, use the [official installation guide](#):

Platform	Installation approach
macOS	Use the macOS package installer or the Homebrew instructions in the official guide.
Linux	Use the appropriate DEB or RPM package and its documented installation steps.

4. Firewall & Network Requirements

4.1 Account-specific endpoints

For corporate networks or VMs, have IT approve the account's service, stage-storage, and certificate-validation endpoints. In Snowsight, run:

```
SELECT SYSTEM$ALLOWLIST();
```

For private connectivity, ask the administrator for the appropriate private endpoint list. Give IT the returned hostnames and ports rather than a generic cloud-storage wildcard list. Most service and file-transfer traffic uses outbound HTTPS on port 443; certificate-validation endpoints can require port 80. Normal client connections do not require an inbound listener.

Installation also requires access to the selected download source. Depending on the installation method, this may include Snowflake's package repository, GitHub, PyPI, or the organization's software mirror. SSO/MFA can require additional identity-provider endpoints.

4.2 Proxy settings

If IT requires a proxy, use its supplied settings. For example, in PowerShell:

```
$env:HTTPS_PROXY = "http://proxy.yourcompany.com:8080"  
$env:HTTP_PROXY = "http://proxy.yourcompany.com:8080"
```

In Command Prompt, use `set HTTPS_PROXY=...` and `set HTTP_PROXY=...`. On macOS/Linux, use `export HTTPS_PROXY=...` and `export HTTP_PROXY=...`. Ask IT which hosts should bypass the proxy through `NO_PROXY`.

For certificate or OCSP errors, have IT check the endpoints, trust chain, proxy, and system clock. Do not disable certificate checks as a setup step. A successful web request alone does not verify login or stage transfers: complete the connection test below and the Day 3 upload check.

5. Named Connection & First Query

5.1 Create the training connection

Run in your terminal:

```
snow connection add -n training
```

Follow the prompts using your account identifier, username, class role, and warehouse. Database and schema can remain unset until the schema setup runs. If training already exists, test it before changing its configuration.

For password authentication, leave the password out of the saved file and supply it through an environment variable when you connect. For configured browser SSO, select `externalbrowser` as the authenticator and leave the password empty. SSO must be available for your account; it is not assumed for individual trial accounts.

5.2 Where settings and credentials are stored

The wizard writes connection settings to a local TOML configuration file. Find your installation's configuration location with:

```
snow --info
```

In the output, find `default_config_file_path`. This is the path to open; you do not need to interpret the other JSON fields. A value of `SNOWFLAKE_HOME: null` by itself is not an error.

If the displayed path is in the standard `%LOCALAPPDATA%\snowflake` folder, open it from Command Prompt with:

```
notepad "%LOCALAPPDATA%\snowflake\config.toml"
```

Windows substitutes your user-specific folder automatically; you do not need to type your username. If `snow --info` shows a different path, open that exact path in Notepad instead. Preserve the existing settings.

If `~/ .snowflake` exists, the default file is `~/ .snowflake/config.toml`. Otherwise the operating-system defaults are:

Platform	Default file
Windows	<code>%USERPROFILE%\AppData\Local\snowflake\config.toml</code>
macOS	<code>~/Library/Application Support/snowflake/config.toml</code>
Linux	<code>~/ .config/snowflake/config.toml</code>

Explicit configuration options and `SNOWFLAKE_HOME` can override the path. Here is an illustrative password-authentication profile with the password omitted:

```
[connections.training]
account = "myorg-myaccount"
user = "myuser"
role = "SYSADMIN"
warehouse = "COMPUTE_WH"
```

Replace the example values with your own. A `password = "..."` entry, if saved, is **plain text**. Do not share credentials in screenshots or lab files. On macOS/Linux, restrict the configuration file to owner read/write permissions.

If `connections.toml` exists in the same directory, CLI reads connections from that file instead. Its section is `[training]`, without the `connections.` prefix. Check which file is active before editing settings.

For password authentication, use PowerShell for the following commands. If you are in Command Prompt, run `powershell` first, then keep using that PowerShell session for the connection tests. PowerShell can prompt without showing the secret in the command or terminal history:

```
$trainingCredential = Get-Credential -UserName 'YOUR_USERNAME' -Message 'Snowflake password'
$env:SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD = $trainingCredential.GetNetworkCredential().Password
```

Replace `YOUR_USERNAME` first. The environment variable contains the password for this terminal session; it is not an encrypted credential store. Clear it when finished, or close the terminal:

```
Remove-Item Env:SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD -ErrorAction SilentlyContinue
Remove-Variable trainingCredential -ErrorAction SilentlyContinue
```

For macOS/Linux Bash, an equivalent hidden prompt is:

```
read -r -s -p "Snowflake password: " SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD
printf '\n'
export SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD
```

Use `unset SNOWFLAKE_CONNECTIONS_TRAINING_PASSWORD` when finished. These are Bash commands; macOS users can start bash first. SSO users skip password variable setup.

5.3 Test the connection

With your password variable set, or SSO configured, run:

```
snow connection test -c training
snow sql -c training -q "SELECT CURRENT_ACCOUNT(), CURRENT_USER(), CURRENT_ROLE(), CURRENT_WAREHOUSE()"
```

Expected: connection status OK, followed by a row showing the intended account, user, role, and warehouse. Approve MFA if prompted. Each CLI invocation can open a new session, so do not assume session context persists between commands.

5.4 File execution and the Day 3 upload check

Run SQL files from the terminal using a quoted path:

```
snow sql -c training -f "path/to/your-script.sql"
```

The path is on the computer where CLI runs. On Day 3, follow Demo 3 in `13-InternalStagesLab.sql` to upload the supplied `movies.csv`. Verify the PUT result and LIST output; connection status OK alone does not prove access to stage-storage endpoints.

6. Lab Files Setup

6.1 Obtain a local copy

Download and extract the instructor's lab package to a writable local folder:

```
Labs/  
  Setup/      Setup scripts and this guide  
  Day1/      Introduction and data loading  
  Day2/      Table types, cloning, Time Travel, SQL, and scripting  
  Day3/      Snowflake CLI, internal stages, Python, and clustering  
  Day4/      Streams, tasks, external tables, and other topics  
  ExtraLabs/  Optional labs
```

A Git-backed Snowsight workspace does not automatically put `movies.csv` on your laptop or VM. Download the Day3 files onto the machine running CLI. Paths can contain spaces; quote file paths as shown in the internal-stages lab.

6.2 Schema setup

Before Day 1, open a SQL file in Snowsight Workspaces and run `Setup/SCHEMA-SETUP-SCOTT.sql` using the instructor-approved role. Verify that `DEMO_DB.SCOTT.DEPT` and `DEMO_DB.SCOTT.EMP` contain data. The CLI connection itself does not create these objects.

6.3 Optional Git setup

If the instructor uses Git integration, run `Setup/GitSetup.sql` with the required administrative privileges, then follow that file's workspace setup instructions. Keep a local copy of files used for CLI transfers.

7. Pre-Installed VM Option

IT should provide a supported operating system, a current browser, a writable student profile, a terminal, and access to the endpoints in Section 4. Install Snowflake CLI using the official platform-specific procedure and verify `snow --version` under the student's login. A separate Python installation is needed for the Python connector demos, with a runtime supported by the chosen connector release.

Use a tested CLI release consistently across class VMs. For automated provisioning, follow the current installer or package-manager documentation; SnowSQL installer switches and paths do not apply to Snowflake CLI.

Pre-install `snowflake-connector-python` in a separate Python virtual environment for `Day3/demo01.py` and `Day3/demo02.py`. Check the current connector requirements before selecting Python. Do not bake personal passwords or authentication caches into the VM image. Students create their own training connection.

8. Verification Checklist

- [] Snowsight login works, including MFA if required.
- [] Account identifier and username are known.
- [] `snow --version` displays the version in a new Command Prompt window on Windows.
- [] The `default_config_file_path` from `snow --info` is understood.
- [] `snow connection test -c training` reports OK.
- [] The test query reports the expected user, role, and warehouse.
- [] The location of the TOML configuration is understood.
- [] IT has approved the required service and stage-storage endpoints.
- [] Lab files, including `Day3's movies.csv`, exist locally and are readable.
- [] `DEMO_DB.SCOTT.EMP` and `DEMO_DB.SCOTT.DEPT` exist and contain data.
- [] If taking the Python demos, the supported Python/connector environment is ready.

9. Troubleshooting

Symptom	Check or action
snow is not recognized	Open a new terminal. Verify installation and PATH with IT.
Named connection not found	Run <code>snow --info</code> ; check the active config file, connection name, and whether <code>connections.toml</code> takes precedence.
Authentication fails	Verify account, username, authenticator, and any required MFA. Password variables apply only to the terminal where they were set.
Browser login fails	Confirm SSO is configured and the terminal can launch/access a browser.
Connection timeout	Check account endpoints, VPN, proxy, and network policy with IT.
Certificate error	Have IT inspect trust/certificate-validation connectivity; do not bypass validation.
Warehouse access fails	Confirm the role has USAGE on the selected warehouse.
PUT fails in Snowsight	Execute the local SQL file through <code>snow sql -c training -f</code>
Local file not found	Use the absolute path on the CLI machine; retain quotes and use forward slashes in the file URI.
Stage not found	Use the lab's database/schema context in the same invocation or fully qualified stage names.
SQL works but upload fails	Check local file permissions and the account-specific stage-storage endpoints.

Optional: encoding warning on Windows

Run this check in Command Prompt:

```
snow helpers detect-encoding
```

If an encoding warning appears, open `config.toml` using the path identified in Section 5.2. Add the following section, preserving all other settings:

```
[cli.encoding]
file_io = "utf-8"
subprocess = "utf-8"
stdout = "utf-8"
```

If `[cli.encoding]` already exists, update its values rather than adding a duplicate section. Save the file. These CLI settings belong in `config.toml`, even when connection profiles are stored in `connections.toml`.

If the diagnostic separately reports that the console is not using UTF-8, run the following in the same console window:

```
chcp 65001
```

This changes the code page for the current console session only.

Repeat the check:

```
snow helpers detect-encoding
```

Expected result:

No encoding issues - your system is properly configured.

This procedure was verified on the instructor's computer with Snowflake CLI 3.27.0. In that test, changing PYTHONUTF8 did not remove the warning; the `config.toml` settings above did. Use the procedure above for this class.

References

- [Snowflake CLI Windows MSI download](#)
- [Snowflake CLI installation](#)
- [CLI configuration](#)
- [Named connections and authentication](#)
- [Executing SQL](#)
- [PUT file paths](#)
- [Account network allowlist](#)
- [Python connector requirements](#)

End of Student Setup Guide